

Genomic Epidemiology Train-the-Trainer Toolkit

Developed by the Chan Zuckerberg Biohub SF Rapid Response Team

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Introduction: Building Sustainable Genomic Epidemiology Capacity

Genomic epidemiology is a vital capability for outbreak detection, transmission tracking, and pathogen surveillance. However, the necessary expertise remains highly centralized, primarily within a small number of high-income, well-resourced institutions. To effectively bridge this global knowledge gap, we must move beyond isolated training events. The goal is to cultivate a self-sustaining network of skilled local trainers capable of adapting, teaching, and passing on this critical knowledge long after external partners depart.

This toolkit was developed by the Chan Zuckerberg Biohub San Francisco's Rapid Response team, synthesizing years of practical experience gained from delivering hands-on training in multiple countries, often in partnership with organizations such as the WHO, Africa CDC, APHL, and Asia PGI. The design of the toolkit centers on one crucial question: *What complete system does a local trainer require to deliver high-quality, independent, consistent, and sustainable genomic epidemiology training?*

The solution is not merely a collection of slides or a single curriculum. It is a structured, modular, and adaptable system grounded in real-world outbreak scenarios. This system equips trainers not only with the content but, crucially, with the pedagogical tools necessary to teach effectively across diverse audiences and resource settings. Every component of this toolkit is engineered for longevity, designed to endure beyond its creators' tenure.

Curriculum Framework & Module Map

Design Principles

Principle	Description
Modularity	Each module is self-contained; trainers can deliver standalone sessions or full sequences
Tiered delivery	Every module has a Core (low-resource, no internet) and an Extended (cloud/command-line) track
Audience-adaptive	Facilitator notes flag content adjustments for lab, epi, and bioinformatics audiences
Tool-agnostic framing	Concepts taught before tools; platform-specific guides are appendices, not core content

Audience Profiles

Profile	Background	Entry Point	Primary Gaps
Lab Scientist	Wet lab, NGS library prep	Module 1	Bioinformatics, phylogenetic interpretation
Epidemiologist/Public Health Officer	Surveillance, outbreak response	Module 2	Sequencing workflow, tree-reading
Bioinformatician	Scripting, pipelines	Module 3	Epi context, public health translation
Mixed cohort	All of the above	Module 1	Shared language, integration across roles

Module Map Overview

TRACK A — CORE (Low-resource, offline-capable)

Module 1 (M1): Foundations of Genomic Epidemiology

Module 2 (M2): Sequencing Technologies & Sample Quality

Module 3 (M3): Introduction to Bioinformatics Concepts

Module 4 (M4): Reference Genomes, Alignment & Variant Calling

Module 5 (M5): Phylogenetics — Building & Reading Trees

Module 6 (M6): Outbreak Investigation Using Genomic Data

Module 7 (M7): Data Sharing, Ethics & Public Health Translation

TRACK B — EXTENDED (Cloud/CLI, internet-dependent)

Module 1 (M1): Foundations of Genomic Epidemiology

Module 2 (M2): Sequencing Technologies & Sample Quality

Module 3 Extended (M3E): Command-Line Bioinformatics (BASH, conda)
Module 4 Extended (M4E): Nextflow/Snakemake Pipelines & Terra/Galaxy
Module 5 Extended (M5E): Augur/Nextstrain & Advanced Phylogenomics
Module 6 Extended (M6E): Real-time Genomic Surveillance Dashboards
Module 8 Extended (M8E): Data Science for Genomic Epi (Python/R)

Track Relationship and Progression

Track A (Core) and Track B (Extended) share the same learning objectives and conceptual foundations. They are not separate curricula; they are two delivery modes of the same curriculum, differentiated by infrastructure and depth.

M1 and M2 are shared foundations for both tracks. All participants, regardless of whether they will follow the Core or Extended path, complete M1 and M2 first. These modules establish the conceptual and laboratory grounding that all subsequent modules build on.

Track A (Core) covers M1–M7 using offline-capable tools, pre-generated datasets, and graphical interfaces (FigTree, Galaxy, etc.). It is designed for settings with limited or intermittent internet, older hardware, or mixed non-computational audiences. All activities can be delivered using printed materials and USB-staged files.

Track B (Extended) extends selected Track A modules (M3→M3E, M4→M4E, M5→M5E, M6→M6E) with cloud- or command-line-based activities, and adds one new module (M8) with no Core equivalent. Extended modules assume completion of their Track A counterpart and introduce additional competencies, hands-on pipeline work, and infrastructure requirements. M7 has no Extended variant — data-sharing and ethics content is the same regardless of the delivery track.

Trainers should assess participants' background and infrastructure before training and select the appropriate track. For mixed cohorts, Core and Extended participants can complete shared concept blocks together; Extended participants then split off to work on pipeline activities, while Core participants work with pre-generated outputs.

Module Specifications (Track A — Core)

M1: Foundations of Genomic Epidemiology

Field	Detail
Duration	3 hours (lecture 1.5h + activity 1.5h)
Prerequisites	None
Audience	All
Learning Objectives	1. Define genomic epidemiology and distinguish it from classical epidemiology.

	2. Explain what genomic data can and cannot answer in an outbreak. 3. Identify key applications: outbreak source attribution, transmission chain reconstruction, AMR/drug resistance surveillance.
Core Concepts	Pathogen evolution, mutation rates, phylogenetic signal, genomics vs genotyping
Key Activity	Case study: "What would you conclude?" — participants interpret 3 outbreak scenarios using only genomic data, then only epi data, then both combined
Assessment	Pre/post quiz (5 conceptual questions)
Facilitator Notes	For epi audiences: emphasise integration with surveillance data. For lab audiences: connect to sequencing output they already produce.
Resource Adaptations	Core: printed case studies + whiteboard phylogeny sketches. Extended: interactive Nextstrain embeds

M2: Sequencing Technologies & Sample Quality

Field	Detail
Duration	4 hours (lecture 2h + hands-on 2h)
Prerequisites	M1
Audience	Lab Scientists (primary), others as context
Learning Objectives	1. Compare Illumina, Oxford Nanopore, and PacBio for public health use cases. 2. Assess sequencing run quality metrics (coverage, Q-score, read length). 3. Identify pre-sequencing sample handling errors and their downstream impact.
Core Concepts	Short-read vs long-read, coverage depth, library prep artefacts, FASTQ format
Key Activity	QC triage exercise: participants receive 6 sequencing run reports and classify as pass/fail/needs-review with justification

Assessment	QC interpretation practical (graded rubric)
Facilitator Notes	Non-lab audiences may find hands-on less relevant — pair them with lab scientists during activity. Emphasise that interpreting QC output is a shared responsibility.
Resource Adaptations	Core: pre-generated QC reports (PDF). Extended: live FastQC/MultiQC on provided datasets

M3: Introduction to Bioinformatics Concepts

Field	Detail
Duration	4 hours
Prerequisites	M1
Audience	All (especially Epi + Lab with no bioinformatics background)
Learning Objectives	1. Describe the genomic analysis pipeline from FASTQ to phylogenetic tree. 2. Interpret key file formats: FASTQ, FASTA, VCF, Newick. 3. Understand what happens "inside the black box" of automated pipelines.
Core Concepts	Pipeline logic, file formats, QC → assembly/alignment → variant calling → phylogenetics
Key Activity	"Follow the data" flowchart exercise — participants trace a synthetic sample through each pipeline step, annotating what is produced and why
Assessment	File format identification quiz + pipeline sequencing exercise
Facilitator Notes	Avoid live demos if internet is unstable. Use static screenshots of Galaxy/Terra with annotated outputs. Bioinformaticians should be assigned as peer mentors during activity.
Resource Adaptations	Core: printed pipeline diagrams + pre-run output files. Extended: Galaxy workflow walkthrough

M4: Reference Genomes, Alignment & Variant Calling

Field	Detail
Duration	5 hours

Prerequisites	M1, M3
Audience	All
Learning Objectives	1. Explain the role of reference genomes in alignment-based analysis. 2. Interpret alignment statistics (% mapped reads, depth, breadth of coverage). 3. Describe how SNPs/indels are called and filtered, and why filtering choices matter.
Core Concepts	Reference bias, mapping vs de novo assembly, SNP vs indel, variant filtering thresholds
Key Activity	Alignment interpretation exercise: compare 3 alignment summaries with different coverage profiles and make consensus genome quality decisions
Assessment	Short-answer case analysis
Facilitator Notes	Emphasise that bioinformatics decisions (thresholds, filters) are scientific decisions — not just software settings. Connect to downstream phylogenetic uncertainty.
Resource Adaptations	Core: pre-computed output summaries. Extended: Minimap2/BWA/iVar on provided datasets

M5: Phylogenetics — Building & Reading Trees

Field	Detail
Duration	6 hours (can split across 2 sessions)
Prerequisites	M1, M3, M4
Audience	All
Learning Objectives	1. Explain key phylogenetic methods: NJ, ML, Bayesian. 2. Build and interpret a basic phylogenetic tree using provided tools. 3. Annotate a tree with metadata and draw epidemiological conclusions. 4. Identify common misinterpretations of phylogenetic trees.
Core Concepts	Tree topology, branch lengths, bootstrap support, clade definition, molecular clock, common ancestor

Key Activity	"What does this tree tell us?" — annotated tree interpretation exercise using real outbreak dataset (anonymised). Participants present conclusions and defend against the facilitator's challenges.
Assessment	Tree interpretation rubric (structured + open-ended)
Facilitator Notes	This is the highest-anxiety module for non-bioinformatics audiences. Normalize uncertainty. Spend extra time on "what trees cannot tell us." FigTree is the recommended Core tool; IQ-TREE + FigTree for Extended.
Resource Adaptations	Core: pre-built trees in FigTree (downloadable). Extended: IQ-TREE + Augur pipeline

M6: Outbreak Investigation Using Genomic Data

Field	Detail
Duration	6 hours
Prerequisites	M1–M5
Audience	All
Learning Objectives	1. Integrate genomic and epidemiological data to reconstruct a transmission network. 2. Distinguish genomic evidence for single vs multiple introduction events. 3. Communicate genomic findings to non-specialist public health decision-makers.
Core Concepts	Introduction events, cluster definition, transmission bottlenecks, genomic + epi concordance/discordance
Key Activity	Full outbreak simulation: teams receive genomic data, case line lists, and travel histories. They must produce a 5-minute briefing to a "ministry official" (played by facilitator).
Assessment	Team presentation rubric + individual reflection
Facilitator Notes	Use a pathogen relevant to your regional context (Lassa, SARS-CoV-2, cholera, mpox, etc.). Rotate team roles (lab lead, epi lead, communicator) to build cross-functional fluency.
Resource Adaptations	Core: printed datasets + whiteboard mapping. Extended: Nextstrain/Microreact for live visualisation

M7: Data Sharing, Ethics & Public Health Translation

Field	Detail
Duration	3 hours
Prerequisites	M1
Audience	All
Learning Objectives	1. Identify major genomic data sharing platforms (GISAID, INSDC, Pathoplexus) and their submission requirements. 2. Describe ethical considerations specific to genomic surveillance data. 3. Translate genomic findings into public health action points.
Core Concepts	Data sovereignty, benefit-sharing, metadata standards, stigmatisation risk, policy translation
Key Activity	"To share or not to share" ethical dilemma discussions (structured small groups) + data submission walkthrough
Assessment	Ethics scenario written response
Facilitator Notes	This module often surfaces strong opinions. Facilitate, don't adjudicate. Use the WHO pathogen access and benefit-sharing framework as a neutral anchor.
Resource Adaptations	Core: printed GISAID submission checklist. Extended: live GISAID/Pathoplexus demo

Module Specifications (Track B — Extended)

Each Extended module assumes the participant has completed the corresponding Track A module. The Track A learning objectives are carried forward and extended, not replaced. Additional competencies, activities, assessments, and infrastructure requirements are specified below.

M3E: Command-Line Bioinformatics (BASH, conda)

Field	Detail
Duration	6 hours
Track A Prerequisite	M3: Introduction to Bioinformatics Concepts
Additional Prerequisites	Basic familiarity with file navigation is helpful but not required
Audience	Bioinformaticians (primary); lab scientists and epi participants with CLI interest

Additional Learning Objectives	1. Navigate a Unix filesystem and manage files using the command line. 2. Create, activate, and manage Conda environments for bioinformatics tool installation. 3. Execute a multi-step bioinformatics pipeline from raw reads to consensus genome. 4. Interpret and troubleshoot common command-line errors.
Extended Concepts	Unix shell (BASH), file paths and permissions, stdin/stdout/stderr, Conda/Mamba environment management, Miniforge, Bioconda, shell scripting basics
Extended Activity	Guided walkthrough: participants process a provided FASTQ dataset through QC (FastQC/MultiQC), trimming (Trimmomatic/fastp), alignment (Minimap2 or BWA), and consensus generation (iVar or Medaka). Final product: a consensus FASTA ready for phylogenetic analysis.
Additional Assessment	Practical: participants receive a broken pipeline command and must diagnose and fix the error; evaluated on accuracy and reasoning
Infrastructure Requirements	Linux or macOS terminal (or WSL2 on Windows); Miniforge installed; internet for initial Conda package downloads (or pre-staged USB); minimum 8 GB RAM, 20 GB free disk
Facilitator Notes	Expect a wide variation in typing speed and comfort with the terminal. Pair fast and slow participants. Resist solving errors — use the prompt 'What does the error message say?'

M4E: Nextflow/Snakemake Pipelines & Terra/Galaxy

Field	Detail
Duration	7 hours (can split across 2 sessions)
Track A Prerequisite	M4: Reference Genomes, Alignment & Variant Calling
Additional Prerequisites	M3E strongly recommended
Audience	Bioinformaticians and advanced lab scientists
Additional Learning Objectives	1. Explain the purpose and structure of workflow management systems (Nextflow/Snakemake). 2. Run nf-core/viralrecon on a provided dataset using Docker or Singularity. 3. Interpret pipeline outputs: MultiQC report, VCF files, and consensus FASTAs. 4. Configure and submit a workflow on Galaxy or Terra using a provided workspace.
Extended Concepts	Workflow management systems, reproducibility and containerization (Docker/Singularity), nf-core pipeline architecture, Terra workspaces and WDL, Galaxy workflow import and execution, pipeline parameter configuration

Extended Activity	(1) Run nf-core/viralrecon on a provided paired-end FASTQ dataset; interpret the MultiQC summary and identify samples failing QC thresholds. (2) Import and run the equivalent workflow on Galaxy or Terra; compare outputs and discuss reproducibility.
Additional Assessment	Written: given a MultiQC report, participants identify which samples pass/fail QC and justify decisions using threshold criteria; Extended: participants modify one pipeline parameter and describe the expected downstream effect
Infrastructure Requirements	Internet required; Docker or Singularity installed; Java 17+ and Nextflow installed; minimum 16 GB RAM recommended; Galaxy account or Terra billing account
Facilitator Notes	Pipeline runs take time — pre-stage completed outputs as a fallback. The comparison between nf-core CLI and Galaxy/Terra should provoke discussion about reproducibility, accessibility, and infrastructure trade-offs — facilitate this, do not resolve it.

M5E: Augur/Nextstrain & Advanced Phylogenomics

Field	Detail
Duration	7 hours (can split across 2 sessions)
Track A Prerequisite	M5: Phylogenetics — Building & Reading Trees
Additional Prerequisites	M3E; M4E recommended
Audience	Bioinformaticians, epi participants with a strong analytical interest
Additional Learning Objectives	1. Build a time-resolved phylogenetic tree using Augur (filter, align, tree, refine, ancestral, export). 2. Annotate and visualise a phylogeny in Auspice with epidemiological metadata. 3. Interpret molecular clock estimates and assess temporal signal in a dataset. 4. Describe the limitations of Bayesian phylogenetics (BEAST) and when it is appropriate vs ML.
Extended Concepts	Augur pipeline steps, Nextstrain JSON format, Auspice metadata mapping, temporal signal assessment (TempEst), molecular clock models, root-to-tip regression, Nextclade clade assignment
Extended Activity	Participants build a complete Nextstrain analysis from consensus FASTAs: filter → align (MAFFT) → build ML tree (IQ-TREE) → refine with molecular clock (TreeTime) → assign clades (Nextclade) → export and visualise in Auspice. Final product: a shareable Auspice JSON with metadata-annotated phylogeny.
Additional Assessment	Practical: participants receive an Auspice JSON with deliberate errors (wrong root, missing metadata field, implausible clock rate) and must identify and describe

	each issue
Infrastructure Requirements	Internet for initial setup; Nextstrain/Augur Conda environment; Auspice (local); IQ-TREE; minimum 8 GB RAM; pre-downloaded Nextstrain/Nextclade datasets for offline fallback
Facilitator Notes	Display the pipeline diagram throughout. Emphasise that each augur subcommand is a scientific choice, not a technical formality. The Auspice visualisation step reliably produces the highest engagement moment in the module.

M6E: Real-time Genomic Surveillance Dashboards

Field	Detail
Duration	5 hours
Track A Prerequisite	M6: Outbreak Investigation Using Genomic Data
Additional Prerequisites	M5E
Audience	Bioinformaticians and public health officers responsible for surveillance systems
Additional Learning Objectives	1. Describe the components of a real-time genomic surveillance system. 2. Configure and update a Nextstrain or Microreact dashboard with new sequence data. 3. Identify the data infrastructure, human capacity, and governance requirements for sustaining a national surveillance system. 4. Evaluate a surveillance dashboard against standard quality criteria.
Extended Concepts	Surveillance system architecture, automated pipeline triggers, dashboard maintenance, data submission cadence, GISAID/Pathoplexus integration, reporting to WHO/regional bodies, sustainability, and handover planning
Extended Activity	(1) Update a live Nextstrain or Microreact dashboard with a new batch of 20 sequences. (2) Evaluate a provided 'surveillance system specification' against a structured rubric covering timeliness, sampling strategy, data sharing, and maintenance plan.
Additional Assessment	Written: surveillance system evaluation rubric; participants must identify at least two gaps and recommend specific improvements
Infrastructure Requirements	Internet required; Nextstrain/Augur environment; Microreact account (free); access to a shared Nextstrain build repository (GitHub); minimum 8 GB RAM
Facilitator Notes	This module surfaces the most important and most difficult conversations in the curriculum — what does it take to sustain a genomic surveillance system after donor funding ends? Do not avoid this. Participants from national reference labs often find this module the most directly relevant to their work.

M8E: Data Science for Genomic Epi (Python/R)

Field	Detail
Duration	8 hours (typically delivered as a 2-day intensive)
Track A Prerequisite	None — M8 has no Core equivalent; requires M3E as minimum prerequisite
Additional Prerequisites	M3E; M4E; basic familiarity with Python or R helpful
Audience	Bioinformaticians and quantitatively oriented epi participants
Additional Learning Objectives	1. Manipulate and clean genomic metadata using Python (pandas) or R (tidyverse). 2. Produce publication-quality visualisations of phylogenetic and epidemiological data. 3. Perform basic genomic epidemiology analyses: SNP distance matrices, cluster detection, and temporal trend analysis. 4. Write reproducible analysis notebooks (Jupyter or R Markdown) that can be shared and re-run.
Extended Concepts	Data wrangling (pandas/tidyverse), SNP distance matrices, ggtree/ete3 for phylogenetic visualisation, epiweeks, reproductive number estimation basics, reproducible research practices, notebook-based analysis
Extended Activity	Participants work through a structured analysis notebook: (1) clean and merge sequence metadata with epi data; (2) compute a pairwise SNP distance matrix and visualise as a heatmap; (3) produce an annotated phylogenetic tree using ggtree (R) or ete3 (Python); (4) export a reproducible HTML report. Participants adapt the notebook to answer one additional question of their choosing.
Additional Assessment	Participants submit their completed and adapted notebook; assessed on code reproducibility, accuracy of analysis, and clarity of the embedded narrative
Infrastructure Requirements	Python 3.10+ with Jupyter and scientific stack (pandas, matplotlib, seaborn, ete3) or R with tidyverse, ggtree, and R Markdown; internet for initial package install; minimum 8 GB RAM; GitHub account recommended for notebook sharing
Facilitator Notes	Language preference (Python vs R) will divide the cohort — provide the notebook in both languages where possible. Pair Python and R users during the adaptation exercise. The most important outcome is comfort with reproducible, documented analysis.

Suggested Delivery Formats

Format	Duration	Modules	Ideal For
Introductory Workshop	3 days	M1, M2, M5 (Core)	Lab scientists new to GenEpi

Epi Integration Workshop	6 days	M1–M7 (Core)	Mixed national public health teams
Bioinformatics Intensives	5 days	M3–M5 + M3E–M5E	Bioinformaticians building pipelines
Genomic Surveillance Intensive	2 days	M6E (requires M5E + M6 completion)	Public health officers managing national surveillance systems
Data Science Intensive	2 days	M8E (requires M3E completion)	Bioinformaticians and quantitative epi staff
Refresher/Update Module	1 day	M5 or M6 (updated dataset)	Previously trained staff

Facilitator Guide Template

This guide is your instructor blueprint for every module in the GenEpi toolkit. Complete one copy per module. Sections marked  are mandatory for all deliveries. Sections marked  are recommended but can be abbreviated for experienced facilitators. Sections marked  are optional enrichment — use them when time and audience allow. Do not skip the Anticipated Challenges or Adaptation Notes sections; these contain hard-won insight from prior deliveries that is difficult to reconstruct independently.

Module Identity

Field	Detail
Module Number & Title	e.g., M5: Phylogenetics — Building & Reading Trees
Toolkit Track	☐ Core (Track A) ☐ Extended (Track B)
Version	e.g., v1.2
Last Updated	
Author(s)	
Reviewed By	
Recommended Audience	☐ Lab Scientists ☐ Epidemiologists/PHOs ☐ Bioinformaticians ☐ Mixed
Prerequisites	List module numbers and any assumed background knowledge
Duration	Total: ___ hrs Lecture: ___ hrs Activity: ___ hrs Assessment: ___ hrs
Maximum Group Size	Recommended: ___ Absolute max: ___
Trainer: Participant Ratio	e.g., 1:15 for hands-on; 1:30 for lecture-only

Learning Objectives

By the end of this module, participants will be able to:

1. *(Remembering/Understanding)* —
2. *(Applying)* —
3. *(Analyzing)* —
4. *(Evaluating/Creating — Extended track only if applicable)* —

Facilitator note: Read these objectives aloud at the start of the session and return to them at the close. Participants should be able to self-assess against each one by the end.

Materials Checklist

Always Required

- ☐ Printed participant handouts (quantity: 1 per participant + 2 spares)
- ☐ Facilitator guide (this document)
- ☐ Pre/post assessment forms
- ☐ Whiteboard or flipchart + markers
- ☐ Timer

Core Track (Low-resource)

- ☐ Pre-printed case study/dataset sheets
- ☐ Pre-generated output files (USB or printed)
- ☐ FigTree installer (USB copy, version: ____)
- ☐ Reference genome file (local copy)

Extended Track (Internet-dependent)

- ☐ Confirmed internet access (test day before)
- ☐ Galaxy/Terra workspace link: ____
- ☐ Conda environment YAML file (filename: ____)
- ☐ Cloud compute credits confirmed

Optional Enrichment

- ☐ Video clips (downloaded offline copies recommended)
- ☐ Guest speaker arranged: ____

Session Timing Plan

Adjust column widths as needed. Add or merge rows for your specific module. Total must match the declared duration above.

Time	Duration	Segment	Facilitator Action	Participant Action	Notes
0:00	10 min	Welcome & objectives	Read objectives aloud;	Complete pre-assessment	Distribute handouts before starting

			administer pre-assessment		
0:10	___ min	Opening hook/framing			
	___ min	Concept block 1			
	___ min	Check understanding for	Pose question; cold-call or pair-share	Respond/discuss	See CFU bank below
	___ min	Concept block 2			
	___ min	Break			Mandatory if session > 2 hrs
	___ min	Hands-on activity	Circulate; observe; prompt, don't solve	Work through the activity	See Activity Guide below
	___ min	Debrief	Facilitate whole-group discussion	Share findings	Use the debrief questions below
	___ min	Synthesis & wrap-up	Return to objectives; summarise key points	Self-assess against objectives	
Last 15 min	15 min	Post-assessment close &	Administer post-assessment; collect; preview next module	Complete post-assessment	

Concept-by-Concept Narrative

For each major concept block in your session, complete one entry below. This is the core of the facilitator guide — write enough detail that a trainer who knows the content but has never taught this module before could deliver it confidently.

Concept Block 1: [Title]

The core idea to convey: *(In 1–2 sentences: what is the single most important thing participants must understand from this block?)*

How to introduce it: *(Opening framing, analogy, or real-world anchor. Specify if different for different audiences.)*

 For lab audiences: ...  For epi audiences: ...  For bioinformatics audiences: ...

Key points to cover:

- Point 1
- Point 2
- Point 3

Suggested analogy or illustration: *(An analogy that works across audience types. Test it — if it doesn't land, have a backup.)*

What to write/draw on the board: *(Sketch or describe the diagram, table, or figure to draw live. Pre-drawn slides are fine, but live drawing often lands better for conceptual content.)*

Check for understanding (CFU):

- Question to pose: "..."
- Expected range of responses:
- How to handle a wrong answer: *(Redirect without embarrassing — e.g., "That's a common intuition — here's where it breaks down...")*
- How to handle no response: *(Pair-share first, then cold call)*

Common misconceptions:

Misconception	Why it arises	How to correct it

Transition to next concept: *(A bridging sentence that connects this concept to the next block.)*

(Duplicate Concept Block section for each additional concept in the module)

Activity Guide

Activity Title:

Purpose: *(What specific learning objective does this activity serve? If it doesn't directly reinforce an LO, cut it.)*

Format: ☐ Individual ☐ Pairs ☐ Small groups (size: ____) ☐ Whole group

Duration: ____ minutes

Instructions to read aloud to participants:

"[Exact or near-exact language to use. Scripting this prevents confusion and saves time]".

What participants receive:

- Item 1
- Item 2

What facilitator does during activity:

- **First 5 minutes:** Circulate and confirm everyone understands the task. Re-read instructions if needed — do not assume confusion means the task is flawed.
- **Middle:** Observe without intervening. Note groups who are stuck vs groups who are finishing early.
- **For stuck groups:** Ask "*What do you know for certain so far?*" — do not provide answers.
- **For fast-finishing groups:** Extension prompt — "*What would change your conclusion?*"

Debrief questions:

1. (*Opening — low stakes, gets everyone talking*)
2. (*Core — directly probes the learning objective*)
3. (*Stretch — pushes toward application or critique*)

What a strong participant response looks like: (*Describe the reasoning, not just the answer. This helps facilitators recognise good thinking even when the specific answer differs.*)

What a struggling response looks like and how to handle it:

Anticipated Challenges

This section captures real difficulties encountered in previous deliveries. Update it after every training.

Challenge	When it typically occurs	Recommended response
Mixed expertise creates uneven pacing	Activity phases	Assign bioinformaticians as peer mentors; give advanced participants extension tasks
Participants challenge the relevance of genomics for resource-limited settings	M1, M6 discussions	Validate the concern; use local outbreak examples; don't dismiss — this is a real tension
Internet/power failure mid-session	Extended track activities	Always have Core track materials as backup; know where the offline dataset copies are
Participants fixate on software rather than concepts	M3, M4, M5	Redirect: " <i>The tool will change. The question you're asking with it won't.</i> "
Tree interpretation causes high anxiety in non-bioinformatics participants.	M5	Normalise: open with " <i>Nobody reads trees fluently the first time.</i> " Use simplified examples first.
(Add from your own experience)		

Adaptation Notes

Adapting for Audience Type

Audience	Emphasize	De-emphasize	Swap or add
Lab Scientists	Wet-lab-to-sequence connection; interpretation QC	Command-line syntax	Pair with M2 content
Epidemiologists/PHOs	Epi integration; public health translation	Pipeline mechanics	Add M6 outbreak framing
Bioinformaticians	Pipeline logic; tool comparisons	Basic biology review	Add Extended track components
Mixed cohort	Shared language; cross-role dependencies	Role-specific deep dives	Assign mixed-role teams for activities

Adapting for Resource Setting

Setting	Adjustments
Low-resource / offline	Use pre-generated output files on USB. Replace live demos with annotated screenshots. Pre-print all datasets. Confirm FigTree installs the day before.
Mid-resource	Galaxy workflows preferred over CLI. Download all datasets at the start of the session, not during. Have offline fallback ready.
High-resource / cloud	Full Extended track available. Ensure Terra/cloud credentials are provisioned ≥ 48 hrs before training.

Adapting for Time Constraints

Available time	Recommended cuts	What must stay
50% of planned time	Cut extended activity; use pre-worked example instead	Core concept blocks + CFUs + post-assessment
75% of planned time	Shorten debrief; reduce group reporting	All concept blocks + at least one activity
Full time	Deliver as planned	—

Localizing Content

- **Substitute the outbreak case study** with a pathogen relevant to your regional context. The dataset structure and questions should remain the same; only the biological context changes.
- **Use local data where possible** — even a single locally-sequenced sample included in a tree dramatically increases engagement.
- **Cite local public health bodies** (national CDC, Ministry of Health) rather than only global organisations when giving policy examples.
- **Flag translation needs** — key terms (phylogenetics, variant calling, lineage) should have agreed local-language equivalents before training begins.

Key Terms Glossary

List the 8–12 most critical terms for this module. Define them in plain language. Provide the printed glossary as a participant handout.

Term	Plain-language definition	Common misuse to flag

Frequently Asked Questions

Document real questions from past trainings. Update after every delivery.

Q: A:

Q: A:

Q: A:

Extended Reading & Resources

Provide 3–5 resources maximum. Annotate each one — an unannotated reading list is rarely used.

Resource	Type	Why it's useful	Access
	Paper / Video / Tool / Guide		Free / Subscription / Offline copy available

Facilitator Self-Evaluation

Complete within 24 hours of delivery. Use findings to update the Anticipated Challenges and Adaptation Notes sections.

1. Which learning objective was most clearly achieved? How do you know?
2. Which objective was least well achieved? What would you change?
3. What question or moment caught you off guard?
4. How well did the timing plan hold? Where did you lose time?
5. What would you tell the next person facilitating this module?

Version & Revision Log

Version	Date	Changed by	Summary of changes
v1.0			Initial release

This template is part of the GenEpi Train-the-Trainer Toolkit developed by Chan Zuckerberg Biohub SF. It may be adapted for local use, provided learning objectives and assessment instruments are retained.

Competency Assessment Tools

Overview of this document: This document contains **participant pre/post knowledge assessment**, which measures learning gain across all Core modules

Participant Pre/Post Knowledge Assessment

Instructions for Facilitators

Administer the **full assessment** at the start of the first module and again at the end of the last module delivered. For standalone single-module workshops, administer at the start and end of that session. Do not review answers between pre and post. Scoring is anonymous — use participant ID codes, not names.

Instructions for Participants

"This assessment helps us understand what the training added to your knowledge — it is not a test of your ability or a gatekeeping exercise. Answer as honestly as you can. Your responses will not affect your participation certificate. Use the ID code on your form, not your name."

Section A: Conceptual Foundations

(Suitable for pre and post)

1. Which of the following best describes the primary added value of genomic data in an outbreak investigation?

- ☐ A. It replaces the need for epidemiological contact tracing
- ☐ B. It provides higher-resolution evidence of transmission relationships than epi data alone
- ☐ C. It confirms the identity of the index case with certainty
- ☐ D. It is only useful after an outbreak has ended

2. A phylogenetic tree shows two patient samples on adjacent branches with very short branch lengths. What is the most defensible interpretation?

- ☐ A. These patients definitely had direct contact
- ☐ B. These patients' viruses are closely related, suggesting possible recent common ancestry
- ☐ C. One patient infected the other
- ☐ D. The samples were collected at the same time

3. A sequencing run returns reads with an average Q-score of 15 and 40% of the reference genome covered at $\geq 10\times$ depth. How should this sample be treated?

- ☐ A. Proceed to phylogenetic analysis — coverage is sufficient
- ☐ B. Flag for repeat sequencing — quality and coverage are below acceptable thresholds
- ☐ C. Use only the high-coverage regions for variant calling
- ☐ D. Q-score is irrelevant for consensus genome generation

4. Which of the following is NOT a limitation of phylogenetic analysis in outbreak investigations?

- ☐ A. Sampling bias can distort tree topology
- ☐ B. Similar sequences may reflect convergent evolution rather than direct transmission
- ☐ C. Phylogenetic trees require epidemiological metadata to be interpretable
- ☐ D. Phylogenetic trees can identify the exact date of transmission events

5. A country sequences 5% of its COVID-19 cases and shares data on GISAID. A global phylogeny places their sequences in a clade with sequences from a neighbouring country. A researcher concludes the outbreak was imported. What is the most important caveat to this conclusion?

- ☐ A. GISAID data is unreliable
- ☐ B. The conclusion may be an artefact of unequal sampling — the true source may be undetected local circulation
- ☐ C. Phylogenies cannot be used for importation analysis
- ☐ D. Neighbouring countries always share the same lineages

Section B: Applied Knowledge

(Best suited for post-assessment; include in pre to establish baseline)

6. You receive a FASTA file with 120 sequences from an outbreak. Which of the following best describes the most important set of quality checks to perform before building a phylogenetic tree?

- ☐ A. Check file size, confirm sequences are in alphabetical order, and verify all sample names are unique
- ☐ B. Assess sequence length consistency, check for excessive ambiguous bases (Ns), and confirm sequences are aligned or alignable
- ☐ C. Confirm the file opens correctly in a text editor, check for duplicate sequences, and verify the reference genome matches the outbreak pathogen
- ☐ D. Run BLAST on all sequences to confirm species identity, remove sequences shorter than 100bp, and check for stop codons

7. A public health official asks: *"The phylogenetic tree shows our outbreak is one cluster — does that mean we have a single source?"* Which response is most accurate and appropriate for a non-specialist audience?

- ☐ A. "Yes — a single cluster confirms there was one introduction event and all cases descend from a common source."
- ☐ B. "Not necessarily. A single cluster means the sequenced viruses are closely related, but we may not have sequenced all cases. We need to combine this with contact tracing and travel data before concluding the source."
- ☐ C. "Phylogenetic trees cannot answer questions about outbreak sources — you would need a different type of analysis for that."
- ☐ D. "A single cluster means either one source or multiple sources — there is no way to distinguish between them using genomic data."

8. You are advising a national reference laboratory with intermittent internet and limited compute on building a genomic surveillance system. Which combination of platform and approach is most appropriate?

- ☐ A. Illumina NovaSeq with a cloud-based Nextflow pipeline — highest accuracy regardless of infrastructure constraints
- ☐ B. Oxford Nanopore MinION with offline-compatible tools (e.g., RAMPART, Medaka, local IQ-TREE) — portable, lower cost, and functional without reliable internet
- ☐ C. PacBio Revio with a Galaxy-based workflow — long reads improve phylogenetic resolution, and Galaxy removes the compute burden
- ☐ D. Any platform is equally suitable — the bioinformatics approach matters more than sequencing technology in resource-limited settings

Section C: Reflection

(Post only)

9. Which of the following best reflects a meaningful change in how a genomic epidemiologist should approach outbreak communication after completing this training?

- ☐ A. Present genomic findings as definitive evidence to strengthen public health messaging and reduce ambiguity for decision-makers
- ☐ B. Reserve genomic findings for peer-reviewed publication before sharing with public health authorities
- ☐ C. Frame genomic findings probabilistically, explicitly acknowledging their limitations, and always integrate them with epidemiological data before drawing conclusions
- ☐ D. Defer all genomic interpretation to bioinformaticians and focus communication on traditional epidemiological findings only

10. Which of the following most accurately describes a gap that genomic epidemiology training alone cannot fully address?

- ☐ A. The ability to build phylogenetic trees from consensus genome sequences
- ☐ B. The structural inequities in data sharing frameworks that limit benefit to low-income countries contributing genomic data
- ☐ C. The interpretation of sequencing quality control metrics
- ☐ D. The differences between short-read and long-read sequencing platforms

Scoring Guide — Section A

Q	Correct	Key distractor to note	What a wrong answer reveals
1	B	A (common among epi audiences)	Misunderstanding of genomics as replacement vs complement
2	B	A or C (over-interpretation of trees)	Conflating genetic relatedness with confirmed transmission
3	B	C (partial use fallacy)	Uncertainty about QC thresholds and downstream consequences

4	D	C (trees do require metadata, so D is the "not a limitation")	Misreading of question logic; review with group if widespread
5	B	A (platform scepticism	Insufficient attention to sampling bias — a critical concept

Section A score: ____ / 5

Scoring Guide — Sections B & C

Q	Correct	Key distractor to note	What a wrong answer reveals
6	B	C (common — conflates file integrity with sequence quality)	Uncertainty about what pre-phylogenetic QC actually involves
7	B	A (over-interpretation of clusters)	Conflating phylogenetic relatedness with confirmed single-source: a critical communication gap
8	B	A (platform preference over context)	Ignoring infrastructure constraints; defaulting to high-resource assumptions
9	C	A (overconfidence in genomic findings)	Misunderstanding the role of uncertainty communication in public health translation
10	B	C or D (technical gaps rather than structural)	Insufficient awareness of equity and data sovereignty dimensions of genomic surveillance

Section B & C score: ____ / 5 Pre-score (Section A only): ____ / 5 Post-score (Sections A + B + C): ____ / 10 Gain: ____

Technical Setup Guides

How to use this document.

This document contains three environment-specific setup guides, each self-contained:

- **Guide A: Offline / Low-Resource Environment** — for labs with no reliable internet or limited compute
- **Guide B: Galaxy & Terra (No-Code / Cloud)** — for participants who will use hosted platforms
- **Guide C: Command-Line Interface (CLI)** — for Linux, macOS, and Windows (WSL2) users

Each guide covers the tools required for the Core and Extended curriculum tracks. Read the **Environment Selection Guide** below first to determine which guide applies to your training cohort.

Environment Selection Guide

Setting	Internet	Compute	Recommended Guide
Field lab, intermittent or no internet	None / unreliable	Basic laptop	Guide A (Offline)
National reference lab, browser access only	Stable	Any	Guide B (Galaxy/Terra)
Bioinformatics training, hands-on pipeline work	Stable	≥8 GB RAM, multicore	Guide C (CLI)
Mixed cohort	Stable	Mixed	Guides B + C in parallel

Use this table to match your training setting to the right setup guide.

Trainer note: If any participant's workstation has less than 8 GB RAM or cannot run Docker/Singularity, redirect that participant to Guide B (Galaxy/Terra) rather than attempting CLI setup. Do not attempt Guide C on locked-down institutional Windows machines that cannot enable WSL2 — use Guide B instead.

Minimum Hardware Requirements

Component	Guide A (Offline)	Guide B (Galaxy/Terra)	Guide C (CLI)
RAM	4 GB	4 GB (browser only)	≥ 8 GB (16 GB recommended)

Storage	32 GB USB + 20 GB local	5 GB local	≥ 50 GB free disk
CPU	Any dual-core	Any	Multicore (4+ cores recommended)
OS	Any with USB boot	Any with a modern browser	Linux, macOS, or Windows with WSL2
Internet	Not required after USB prep	Required (hosted)	Required for initial setup only

Version Reference (Verified June 2026)

Re-verify these version numbers before each training delivery — tool versions update frequently.

Tool	Current Version	Notes
Miniforge	3 (latest)	Replaces deprecated Mambaforge
Augur (Nextstrain)	33.4.1	PyPI: nextstrain-augur
Nextstrain CLI	10.4.2	
IQ-TREE	3.1.3	Executable: iqtree3
FigTree	1.4.4	Java-based; see macOS note
Nextflow	25.10.4	Requires Java 17+
nf-core/tools	4.0.2	Requires Python 3.10+
nf-core/viralrecon	3.0.0	
nf-core/fetchngs	1.12.0	
nf-core/phyloplace	2.0.0	

Guide A: Offline / Low-Resource Environment

Overview

This guide prepares a USB drive that contains everything needed to run Core track bioinformatics exercises without internet access. It is designed for trainers preparing materials **before** travel to a low-resource setting. All downloads occur once on a machine with internet access; participants then install from the USB.

What Can Run Fully Offline

Tool	Offline capable	Notes
IQ-TREE 3	Yes	Standalone binary, no install needed
FigTree 1.4.4	Yes	Requires Java pre-installed
Augur (core functions)	Yes	Once Conda environment is staged
Nextclade	Yes	Once datasets are pre-downloaded
Auspice (local viewer)	Yes	Runs on local Node.js server
Nextflow + nf-core pipelines	Yes	Once Singularity images + references are cached
Galaxy / Terra	No	Hosted platforms require internet

Part A1: Prepare the USB Drive (Trainer Task — Requires Internet)

Do this **at least 48 hours before** training to allow time for large downloads.

USB Structure

```
□ /GenEpi_Toolkit_USB/  
├─ installers/  
|   ├─ Miniforge3-Linux-x86_64.sh  
|   ├─ Miniforge3-MacOSX-arm64.sh  
|   ├─ Miniforge3-MacOSX-x86_64.sh  
|   ├─ Miniforge3-Windows-x86_64.exe  
|   ├─ OpenJDK17_Linux.tar.gz  
|   └─ OpenJDK17_macOS.pkg
```

```

|   └─ OpenJDK17_Windows.msi
|   └─ nextflow                ← self-installing Nextflow binary
└─ binaries/
    |   └─ iqtree3-Linux/
    |   └─ iqtree3-macOS/
    |   └─ iqtree3-Windows/
    |   └─ FigTree_v1.4.4/
└─ conda_packages/
    |   └─ nextstrain_env.tar.gz    ← packed with conda-pack
└─ nf_pipelines/
    |   └─ viralrecon_singularity.tar.gz ← downloaded with nf-core pipelines download
    |   └─ references/
    |       └─ MN908947.3.fasta      ← SARS-CoV-2 reference
    |       └─ primers/
└─ nextclade_datasets/
    |   └─ sars-cov-2/
└─ training_data/
    |   └─ module_3_exercise/
    |   └─ module_4_exercise/
    |   └─ module_5_exercise/
    |   └─ module_6_simulation/
└─ README.txt

```

Step 1: Download Miniforge installers

❏ # Linux

```
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-Linux-x86_64.sh"
```

macOS Intel

```
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-MacOSX-x86_64.sh"
```

```
# macOS Apple Silicon
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-MacOSX-arm64.sh"

# Windows – download Miniforge3-Windows-x86_64.exe from:
# https://github.com/conda-forge/miniforge/releases/latest
```

Step 2: Download IQ-TREE 3 binaries

Download the appropriate archive for each OS from:
<https://github.com/iqtree/iqtree2/releases/tag/v3.1.3>

- `iqtree-3.1.3-Linux-intel.tar.gz`
- `iqtree-3.1.3-macOS-universal.tar.gz`
- `iqtree-3.1.3-Windows.zip`

Extract and place each in the `/binaries/` folder on the USB.

Step 3: Download FigTree

Download `FigTree_v1.4.4.tgz` (cross-platform Java archive) from:
<https://github.com/rambaut/figtree/releases/tag/v1.4.4>

Use the `.tgz` cross-platform build rather than the macOS `.dmg` — it avoids the known Java detection failure on macOS.

Step 4: Build and pack the Nextstrain Conda environment

On a Linux or macOS machine with Miniforge installed:

```
❑ # Create the environment
conda create -n nextstrain -c conda-forge -c bioconda \
  --override-channels --strict-channel-priority \
  augur auspice nextclade snakemake \
  mafft raxml fasttree iqtree \
  seqkit vcftools sqllite -y
# Pack it for offline transport
conda install -c conda-forge conda-pack -y
conda pack -n nextstrain -o nextstrain_env.tar.gz
```

❑ Copy `nextstrain_env.tar.gz` to `/conda_packages/` on the USB.

Step 5: Download nf-core/viralrecon with Singularity images

```
❑ pip install nf-core
nf-core pipelines download viralrecon \
  --revision 3.0.0 \
```

```
--container-system singularity \  
--compress tar.gz \  
--outdir viralrecon_singularity
```

□ Copy the resulting `.tar.gz` to `/nf_pipelines/` on the USB.

Step 6: Download Nextclade datasets

```
□ nextclade dataset get \  
  --name sars-cov-2 \  
  --output-dir nextclade_datasets/sars-cov-2
```

Step 7: Download reference genome and primers

```
□ # SARS-CoV-2 reference (MN908947.3)  
curl -O  
"https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=nucleotide&id=MN908947.3&ret  
type=fasta&retmode=text" \  
  -o MN908947.3.fasta
```

□ ARTIC primer schemes are available at: <https://github.com/artic-network/primer-schemes>

Part A2: Participant Installation from USB

Install Miniforge

Linux / macOS:

```
□ bash /path/to/usb/installers/Miniforge3-$(uname)-$(uname -m).sh  
# Follow prompts; allow conda to initialize your shell  
source ~/.bashrc    # or ~/.zshrc on macOS
```

□ **Windows:** Run `Miniforge3-Windows-x86_64.exe` and follow the installer—open "Miniforge Prompt" from the Start Menu for all subsequent commands.

Install IQ-TREE 3

Linux:

```
□ tar -xzf /path/to/usb/binaries/iqtree-3.1.3-Linux-intel.tar.gz  
echo 'export PATH="$HOME/iqtree-3.1.3-Linux-intel/bin:$PATH"' >> ~/.bashrc  
source ~/.bashrc  
iqtree3 --version    # verify
```

□ **macOS:**

```

❑ tar -xzf /path/to/usb/binaries/iqtree-3.1.3-macOS-universal.tar.gz
echo 'export PATH="$HOME/iqtree-3.1.3-macOS-universal/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
iqtree3 --version

```

❑ **Windows:** Extract the .zip file, open Command Prompt in the bin folder, and run iqtree3.exe --version.

Install FigTree

```

❑ # All platforms - requires Java 8+
tar -xzf /path/to/usb/binaries/FigTree_v1.4.4.tgz
cd FigTree_v1.4.4
# Launch:
java -jar lib/figtree.jar

```

❑ If Java is not installed, use the OpenJDK 17 installer from the USB /installers/ folder first.

Restore the Nextstrain Conda environment

```

❑ mkdir -p $HOME/envs/nextstrain
tar -xzf /path/to/usb/conda_packages/nextstrain_env.tar.gz \
  -C $HOME/envs/nextstrain
source $HOME/envs/nextstrain/bin/activate
conda-unpack

```

Verify all tools

```

❑ augur --version
nextclade --version
iqtree3 --version
snakemake --version

```

Part A3: Offline Troubleshooting

Problem	Cause	Fix
conda: command not found	Shell not initialised	Run <code>source ~/.bashrc</code> or reopen terminal
conda-unpack: command not found	conda-pack not in environment	Run <code>pip install conda-pack</code> first (need internet) or re-stage USB
FigTree won't open on macOS	Java not found	Install OpenJDK 17 from USB; launch via <code>java -jar</code>

IQ-TREE permission denied	Binary not executable	Run <code>chmod +x iqtree3</code>
Nextflow fails without internet	Plugin fetch attempted	Set <code>NXF_OFFLINE=TRUE</code> and ensure Singularity cache is present

Guide B: Galaxy & Terra (No-Code / Cloud)

Overview

This guide covers setup for two hosted bioinformatics platforms that require no local software installation and run entirely in a web browser. Galaxy is recommended for most training contexts; Terra is recommended when participants need to run production WDL workflows at scale.

Part B1: Galaxy Setup

Choosing a Galaxy Server

Server	URL	Region	Default Quota	Best for
usegalaxy.org	https://usegalaxy.org	Global (US-hosted)	250 GB	Default choice
usegalaxy.eu	https://usegalaxy.eu	Europe	250 GB	EU-based participants
usegalaxy.org.au	https://usegalaxy.org.au	Australia/Pacific	100 GB (600 GB with AU affiliation)	APAC-based participants

All three servers are free, maintained by the Global Galaxy Project, and updated within 90 business days of each Galaxy release. For training, any server will work — pick the one geographically closest to your participants to minimise upload latency.

Registering an Account

1. Go to your chosen server URL
2. Click **Login or Register** → **Register here**
3. Enter your email, a username, and a password
4. Check your inbox for a confirmation email and click the link
 - If the email does not arrive within 5 minutes, check spam; if still absent, use a different email address rather than registering again (duplicate accounts may be suspended)
5. Log in — you will see the Galaxy interface with the tool panel on the left and history panel on the right

Trainer note: Ask participants to register accounts **at least 24 hours before** training. Confirmation emails are occasionally delayed, and account activation is required to upload data and run workflows.

Importing Training Workflows

Recommended workflows for viral genomics training are maintained by the Intergalactic Workflow Commission (IWC) and tagged covid19.galaxyproject.org.

To import a workflow:

1. In Galaxy, click **Workflow** in the top menu
2. Click **Import** (top right)
3. Select the **GA4GH servers (TRS)** tab
4. Search for **sars-cov-2** with organisation **iwc**
5. Select the appropriate workflow (e.g., *SARS-CoV-2 variant calling from Illumina paired-end data*) and click **Import**

Alternatively, import directly from WorkflowHub:

- Go to <https://workflowhub.eu> and search for **sars-cov-2 IWC**
- Click the workflow → **Run in Galaxy** → select your server

Available IWC SARS-CoV-2 Workflows

Workflow	Data type	Use in curriculum
Variant calling — Illumina paired-end	Illumina	M4, M6
Variant calling — Illumina single-end	Illumina	M4
Variant calling — Nanopore (ARTIC)	ONT	M4 Extended
Consensus + Pangolin lineage assignment	Illumina/ONT	M6
Phylogenetics (IQ-TREE + Newick)	FASTA	M5

Uploading Training Data

1. Click the **Upload** icon (arrow up) in the tool panel
2. For small files: drag and drop or click **Choose local file**
3. For data from URLs: click **Paste/Fetch Data** and paste a direct download URL
4. For SRA accessions: use the **Get Data → Download and Extract Reads in FASTQ** tool, entering SRA accession numbers directly

Storage Management

- Galaxy histories are stored in your account quota (250 GB default on org/eu)
- Delete intermediate datasets during training to conserve space: click the dataset → **Delete**
- Permanently purge deleted datasets: **User → Preferences → Manage your storage → Purge deleted datasets**
- For large training cohorts, consider using **Galaxy Training Network (GTN) sandboxes** which provide temporary accounts with pre-loaded data — contact the GTN team at <https://training.galaxyproject.org/training-material/about.html>

Pre-Loading a Shared History for Training

To give all participants the same starting data:

1. Upload all training datasets to your account and run any pre-processing steps
 2. Click **History options** → **Share or Publish**
 3. Select **Make History Accessible** and copy the link
 4. Share the link with participants — they import it via **User** → **Histories** → **Import from File/URL**
-

Part B2: Terra Setup

Note: Terra uses real Google Cloud infrastructure. Costs accrue for compute and storage beyond the free credit period. Do not run large analyses without understanding the cost implications.

Account Registration

1. Go to <https://app.terra.bio>
2. Click **Sign in with Google** — use a personal Google account (work/institutional Google accounts tied to an organisation's GCP are usually ineligible for the \$300 free credit)
3. Accept the Terra Terms of Service
4. New users receive **\$300 in free Google Cloud credits**, valid for 90 days — sufficient for testing and small training exercises.

Trainer note: The \$300 credit covers tutorial-scale analyses (typically under \$1–\$20 per workflow run) but is unlikely to cover full-scale genomic analyses. If participants are using institutional accounts, verify credit eligibility before training.

Setting Up a Billing Project

A Billing Project is required to create workspaces and run workflows.

1. Go to the **Google Cloud Console** (<https://console.cloud.google.com>) using the same Google account as Terra.
2. Navigate to **Billing** → **Create a billing account** and link a payment method (the \$300 credit activates here)
3. In the Cloud Console IAM settings, grant the service account terra-billing@terra.bio the **Billing Account User** role.
4. Return to Terra → **Billing** → **Add Billing Project** → link to your GCP billing account

Creating a Workspace

1. In Terra, click **Workspaces** → **Create a New Workspace**
2. Give it a name (e.g., [genepi-training-2026](#))
3. Select your Billing Project
4. Set authorisation domain if needed (leave blank for open training)

Importing Workflows from Dockstore (Theiagen PHB)

Theiagen's Public Health Bioinformatics (PHB) workflows are the recommended viral genomics workflows for Terra-based training.

1. Go to <https://dockstore.org> and search for "TheiaCoV" or "Theiagen Public Health Bioinformatics".
2. Select a workflow (e.g., [TheiaCoV_Illumina_PE](#))
3. On the workflow page, click **Launch with Terra** (select WDL)
4. Choose your destination workspace and click **Import**
5. In Terra, go to your workspace → **Workflows** tab → find the imported workflow
6. Upload a JSON inputs file in the **Inputs** tab (test parameter files are not imported automatically — download them from the workflow's Dockstore page)

Recommended Theiagen PHB Workflows for Training

Workflow	Purpose	Module
TheiaCoV_Illumina_PE	Illumina paired-end viral genome assembly + QC	M4, M6
TheiaCoV_ONT	Nanopore viral genome assembly + QC	M4 Extended
TheiaProk_Illumina_PE	Bacterial genome assembly (AMR)	Optional
Snippy_Variants	SNP calling vs reference	M4

Uploading Data to Terra

1. In your workspace, go to the **Data** tab
2. Click **Files** → **Upload** to add FASTQ files or reference genomes
3. For large datasets, use [gsutil](#) (Google Cloud Storage command-line tool):

```
gsutil cp *.fastq.gz gs://your-workspace-bucket/data/
```

□ The workspace bucket URL is found under **Data** → **Files** → **Bucket URL**.

Cost Management

- Stop virtual machines when not in use: **Analysis** → **Environments** → **Stop**
- Delete intermediate output files after each exercise
- Monitor spend: **Billing** → **View GCP billing** in the Cloud Console
- For training, use the smallest available VM (e.g., [n1-standard-2](#)) and avoid GPUs

Guide C: Command-Line Interface (CLI)

Overview

This guide covers installation of all CLI tools for the Extended curriculum track on Linux (Ubuntu), macOS, and Windows (via WSL2). Follow the **OS-specific prerequisites** for your platform first, then proceed to the shared tool installation steps.

Part C0: OS-Specific Prerequisites

Linux (Ubuntu 22.04 LTS or later)

```
# Update system packages
sudo apt update && sudo apt upgrade -y

# Install essential build tools and utilities
sudo apt install -y curl wget git unzip build-essential \
    libssl-dev zlib1g-dev libbz2-dev libreadline-dev \
    libsqlite3-dev libffi-dev

# Verify
curl --version && git --version
```

macOS (Monterey 12 or later)

```
# Install Xcode Command Line Tools
xcode-select --install

# Install Homebrew (package manager)
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

# Add Homebrew to PATH (Apple Silicon)
echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zshrc
source ~/.zshrc

# Verify
brew --version
```

Windows — Enable WSL2 (Required)

Almost all tools in this guide are not natively supported on Windows PowerShell. WSL2 with Ubuntu is required. The only exceptions are FigTree (Java GUI) and IQ-TREE (Windows binary), which can run natively.

1. Open **PowerShell as Administrator** and run:

❑ `wsl --install`

❑ This installs WSL2 with Ubuntu by default. Restart when prompted.

2. On restart, Ubuntu will complete setup— prompting you to create a Unix username and password.
3. Install **Docker Desktop** from <https://www.docker.com/products/docker-desktop>.
 - During install, ensure **Use WSL2 instead of Hyper-V** is selected
 - After install: Docker Desktop → Settings → Resources → WSL Integration → enable your Ubuntu distro
4. All subsequent CLI steps in this guide should be run inside the **Ubuntu WSL2 terminal**, not PowerShell

Part C1: Conda / Mamba (Miniforge)

Install Miniforge, not Anaconda. Mambaforge was deprecated in January 2025.

Linux / macOS

❑ # Download the correct installer for your platform

```
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-$(uname)-$(uname -m).sh"
```

Install (follow prompts; accept license; allow conda to initialize your shell)

```
bash Miniforge3-$(uname)-$(uname -m).sh
```

Reload shell

```
source ~/.bashrc    # Linux
```

```
source ~/.zshrc     # macOS
```

Verify

```
conda --version
```

```
mamba --version
```

Windows (inside Ubuntu WSL2)

Follow the Linux steps above inside the WSL2 Ubuntu terminal.

Configure Bioconda channels (do once)

❑ `conda config --add channels bioconda`

```
conda config --add channels conda-forge
```

```
conda config --set channel_priority strict
```

❑ **Important:** Never install bioinformatics tools directly into the `base` environment. Always create named environments (see below).

Part C2: Nextstrain (Augur + Nextstrain CLI)

Create a Nextstrain environment

```
❑ conda create -n nextstrain \
  -c conda-forge -c bioconda \
  --override-channels --strict-channel-priority \
  augur auspice nextclade snakemake git \
  mafft raxml fasttree iqtree \
  seqkit vcftools epiweeks ncbi-datasets-cli \
  --yes
conda activate nextstrain
```

Install Nextstrain CLI

```
❑ # Inside the nextstrain environment
pip install nextstrain-cli
# Set up the Conda runtime (recommended for training)
nextstrain setup conda
# Verify
nextstrain check-setup
augur --version
nextclade --version
```

Download a Nextclade dataset (SARS-CoV-2 example)

```
❑ nextclade dataset get \
  --name sars-cov-2 \
  --output-dir nextclade_datasets/sars-cov-2
```

Common Nextstrain workflow test

```
❑ # Run the Nextstrain Zika tutorial to verify the full stack
nextstrain build https://github.com/nextstrain/zika --cpus 2
❑
```

Part C3: IQ-TREE 3

IQ-TREE runs as a standalone binary — no Conda environment needed.

Linux

```
❑ wget https://github.com/iqtree/iqtree2/releases/download/v3.1.3/iqtree-3.1.3-Linux-
intel.tar.gz
```

```
tar -xzf iqtrees-3.1.3-Linux-intel.tar.gz
mkdir -p $HOME/.local/bin
cp iqtrees-3.1.3-Linux-intel/bin/iqtrees3 $HOME/.local/bin/
echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc
iqtrees3 --version
```

macOS

```
❏ curl -LO https://github.com/iqtrees/iqtrees2/releases/download/v3.1.3/iqtrees-3.1.3-macOS-universal.tar.gz
tar -xzf iqtrees-3.1.3-macOS-universal.tar.gz
mkdir -p $HOME/.local/bin
cp iqtrees-3.1.3-macOS-universal/bin/iqtrees3 $HOME/.local/bin/
echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
iqtrees3 --version
```

Windows (native binary — no WSL2 needed)

1. Download `iqtrees-3.1.3-Windows.zip` from <https://github.com/iqtrees/iqtrees2/releases/tag/v3.1.3>
 2. Extract to `C:\Tools\iqtrees3\`.
 3. Add `C:\Tools\iqtrees3\bin` to your system PATH via **System Properties → Environment Variables**
 4. Open Command Prompt and run `iqtrees3.exe --version`
-

Part C4: FigTree

FigTree is a Java GUI application — it runs natively on all platforms.

Install Java 17+ (if not already installed)

Linux:

```
❏ sudo apt install -y openjdk-17-jre
java -version
```

macOS:

```
❏ brew install --cask temurin@17
java -version
```

❑ **Windows:** Download and install Temurin 17 from <https://adoptium.net>

Install FigTree

Download FigTree_v1.4.4.tgz from: <https://github.com/rambaut/figtree/releases/tag/v1.4.4>

```
# Linux / macOS
tar -xzf FigTree_v1.4.4.tgz
cd FigTree_v1.4.4
# Launch
java -jar lib/figtree.jar
```

❑ **Windows:** Extract the .zip version, navigate to the lib/ folder, and run:

❑ `java -jar figtree.jar`

❑ **macOS known issue:** The .dmg installer frequently fails with "No suitable Java version found" on macOS 13+ and Apple Silicon, even when Java is installed. Use the .tgz cross-platform build and launch via `java -jar` as shown above.

Part C5: Nextflow + nf-core

Install Java 17+

Nextflow requires Java 17 or later (up to Java 26). Use SDKMAN! for the most reliable cross-platform Java management.

```
❑ # Install SDKMAN!
curl -s "https://get.sdkman.io" | bash
source "$HOME/.sdkman/bin/sdkman-init.sh"
# Install Temurin Java 17 LTS
sdk install java 17.0.11-tem
java -version # should show 17.x
```

Install Nextflow

Do not use Conda to install Nextflow — it frequently causes version and Java compatibility issues. Use the official self-installing script.

```
❑ curl -s https://get.nextflow.io | bash
mkdir -p $HOME/.local/bin
mv nextflow $HOME/.local/bin/
echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.bashrc # Linux
# or ~/.zshrc on macOS
```

```
source ~/.bashrc
nextflow -version
```

Install nf-core tools

```
# Create a dedicated environment
conda create -n nf-core -c bioconda -c conda-forge nf-core python=3.12 --yes
conda activate nf-core
nf-core --version
```

Run a test pipeline (requires internet + Docker or Singularity)

```
# Test nf-core/viralrecon with the built-in test profile
nextflow run nf-core/viralrecon \
  -revision 3.0.0 \
  -profile test,docker \
  --outdir viralrecon_test_output
```

Download a pipeline for offline use

```
conda activate nf-core
nf-core pipelines download viralrecon \
  --revision 3.0.0 \
  --container-system singularity \
  --compress tar.gz \
  --outdir viralrecon_offline
```

Set environment variables for offline / HPC use

```
# Add to ~/.bashrc or ~/.zshrc
export NXF_SINGULARITY_CACHEDIR="$HOME/singularity_cache"
export NXF_OFFLINE=TRUE # prevents pipeline auto-updates
export NXF_HOME="$HOME/.nextflow"

```

Part C6: Full Environment Verification

Run this checklist after completing all installations. Every command should return a version number without errors.

```
# Activate the Nextstrain environment
conda activate nextstrain
augur --version # e.g. augur 33.4.1
```

```

nextclade --version      # e.g. nextclade 3.x.x
snakemake --version
mafft --version
iqtree3 --version        # e.g. IQ-TREE 3.1.3
# Switch to nf-core environment
conda activate nf-core
nextflow -version         # e.g. 25.10.4
nf-core --version         # e.g. 4.0.2
# Standalone (not in an environment)
java -version             # must show 17.x or higher
java -jar path/to/figtree.jar # FigTree GUI should open

```

□

Part C7: CLI Troubleshooting

Problem	Platform	Cause	Fix
conda: command not found	All	Shell not initialised	Run <code>source ~/.bashrc</code> or reopen terminal
augur: command not found	All	Wrong environment active	Run <code>conda activate nextstrain</code>
mamba: command not found	All	Miniforge not installed (installed Anaconda instead)	Reinstall with Miniforge3
Nextflow fails: "Java 17 required"	All	Old Java version	Install Temurin 17 via SDKMAN!
nf-core pipelines download not found	All	Old nf-core version	nf-core 4.0+ uses <code>pipelines</code> prefix: <code>nf-core pipelines download</code>
Docker permission denied	Linux	User not in docker group	<code>sudo usermod -aG docker \$USER</code> then log out and back in
FigTree: "No suitable Java"	macOS	macOS Java detection bug	Use <code>java -jar lib/figtree.jar</code> instead of double-clicking
WSL2 not available	Windows	Virtualisation disabled in BIOS	Enable virtualisation in BIOS/UEFI firmware settings

IQ-TREE: "cannot execute binary"	Linux	Wrong architecture binary	Download the Linux-intel build for x86_64; Linux-aarch64 for ARM
NXF_OFFLINE still fetching	All	Plugin cache missing	Run nextflow plugin install once online, then set NXF_OFFLINE=TRUE

Version & Revision Log

Version	Date	Changed by	Summary
v1.0	August 2026	CZ Biohub SF GenEpi Team	Initial release — verified against tool versions as of June 2026

This document is part of the GenEpi Train-the-Trainer Toolkit developed by Chan Zuckerberg Biohub SF. Version numbers and URLs should be verified before each training delivery — bioinformatics tools update frequently.